



$$\frac{1}{4}n^2\langle\sigma v\rangle E_\alpha = \frac{3nT}{\tau_E}$$

$$\Rightarrow n\tau_E = \frac{12T}{\langle\sigma v\rangle E_\alpha} \quad \Rightarrow \quad nT\tau_E \gtrsim \text{const}$$

three steps, no new physics: cancel one power of n , solve for $n\tau_E$, then multiply by T to reach the form the reactor literature quotes

Explanatory sketch; schematic, not to scale.